



Design of near-infrared solid-state tunable Fabry-Perot filters based on VO₂/P4VP films

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Abstract: The significant refractive index change of VO₂ during phase transition makes it attractive for the design of optical switches and filters, but high absorption of VO₂ limits its development in the field of filter. In this paper, we firstly design near-infrared solid-state tunable/switchable single-/dual-band bandpass Fabry-Perot (F-P) filters based on VO₂/P4VP composite films using COMSOL software. The simulation results show that the single-passband tunable filter can achieve a continuous thermal tuning range of 19 nm at 1064 nm, the full width at half maxima (FWHM) is 28.8 nm, and the peak transmittance is >30%. On this basis, we simulated a tunable dual-band bandpass filter, the left peak can be heated to achieve a tuning range of 15 nm (945 nm), the right peak is 24 nm (1168 nm), the peak transmittances of both the two peaks are >35%. When boosting the VO₂ fraction, the continuous tunability of the above-mentioned single-/dual-band filters will be switchable due to the high absorption of VO₂, and then switchable single-/dual-band bandpass filters are designed. This work has guiding significance for the design of novel filter structures and optical tunable devices based on phase change materials such as VO₂ in the future.

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1. Introduction

In 1959, F. J. Morin [1] discovered that VO₂ underwent an insulator-metal phase transition near T = 68 °C. Since then, VO₂ has gradually become a focus of attention in the field of condensed matter physics and phase change materials. The insulator-metal phase transition is accompanied by the transformation of the lattice structure (monoclinic phase to tetragonal rutile phase) with dramatic changes in physical properties such as electrical conductivity, refractive index, optical absorption, etc [2–6]. Therefore, vanadium dioxide is attracting attention in the field of wavelength/frequency selective optoelectronic devices, such as optical switches and tunable filters.

In 2008, Givernaud et al [7]. first reported tunable stopband filters operating at 11–13 GHz based on the integration of VO₂ switches. Since then, the tuning effect of VO₂ in traditional GHz and THz bands has been widely studied and progressed. VO₂ also shows the potential of tuning filtering in the near-infrared band. For example, Boyce et al [5]. integrated VO₂ with plasmonic nanogap cavities between gold arrays and a gold layer to achieve dynamic tuning of absorption resonances in the near-infrared band (1340–1300 nm). **Table 1** summarizes the research progress in VO₂-based tunable filter in recent years. Currently, VO₂-based filters have a singular structure and are progressing at a slow pace within the realm of transmissive infrared narrowband filtering.

The solid-state F-P filter structure is a stable and efficient filter structure with great development potential and commercial value by changing the refractive index of the cavity layer material

Table 1. Research progress of VO₂-based tunable filters.

Date	Structure	Filter type	Tuning type	Tuning range	FWHM or Bandwidth	Ref.
2009	Metamaterial	Reflective narrowband bandpass	Thermal	3.0-3.1 μm	0.7 μm at 3.0 μm	[28]
2013	Metamaterial	Transmissive Dual-band bandpass	Thermal	30 GHz	82.0 GHz at 0.41 THz	[29]
2016	Bragg gratings	Reflective bandpass	Thermal	742-795 nm	-	[8]
2018	DGS shape	Bandstop	Thermal	28.2-35 GHz	-	[30]
2019	Metasurface	Transmissive bandpass	Thermal	400 nm at 1.75 μm	-	[31]
2019	Metamaterial	Transmissive bandpass	Electrical	-	-	[32]
2021	Multilayer	Reflective Bandstop	Thermal	1906-2960 nm	-	[33]
2021	Nano plasmonic	Transmissive bandpass	Size	650 nm	-	[34]
2022	Multilayer	Transmissive narrowband bandpass	Thermal	Narrowband (at 8.8 μm) to Broadband (8–12 μm).	-	[6]
2022	Metasurface	Transmissive narrowband bandpass	Size and Thermal	0.88-0.81 THz (y-polarized THz field)	-	[35]
2022	Asymmetric F-P cavity	Reflective Dual-band bandpass	Thermal	500 & 700 nm-480 & 640 nm	-	[36]
2023	Interdigitated electrodes	-	Thermal	733-713 MHz	24 MHz	[37]

to achieve a tunable filter output. In 2016, Xi Wang et al [8]. at the University of California, Berkeley, sandwiched a VO₂ defect layer in the middle of the SiO₂/TiO₂ Bragg gratings, and its temperature-dependent reflectance spectra showed the dynamic tunability of the Bragg filter, which provided an idea for the combination of traditional phase change materials and optical cavities. Subsequently, Xie et al [9]. realized an electrically tunable infrared (IR) narrow-bandwidth filter with a transmittance of ~6% in the range of 3 ~ 5 μm by filling the F-P microcavity with nematic liquid crystal material. In 2018, Ranjan Singh et al [10]. designed a tunable narrow-band absorption filter operating in the visible band using the thermal phase change material Ge₂Sb₂Te₅ (GST) as an asymmetric F-P cavity material. Similarly, Calum Williams et al [11]. at the University of Cambridge used the distributed Bragg reflector (DBR)-GST-DBR structure to implement a narrowband filter with 4.5 μm and 4.75 μm center wavelength switching. In 2023, in order to solve the limitation of transmittance by a single cavity and achieve narrowband at the same time, Zhang's team [12] proposed a narrowband tunable electro-optical filter based on dual F-P cavities by using lithium niobate (LiNbO₃, LN) and MgF₂ as film pairs. The filter has a high transmittance of 99.9% and a narrow passband of about 0.3 nm. When the voltage is increased from -150 V to 150 V, the resonance wavelength around 1550 nm can be blue shifted by 102 nm. However, the solid-state tunable F-P filter currently has some shortcomings, such as complex preparation process, low transmittance, single working band, and poor continuous tuning, which limits its development.

VO₂ has significant advantages in the design of solid-state F-P filters, but its strong absorption will lead to propagation losses, which will affect filter performance. Compared with pure VO₂ film, VO₂/polymer film has lower absorption, exhibits higher transmittance and retains the thermochromic characteristics of the composite films [13,14]. Among many polymers, P4VP has a high melting point, exhibits good affinity for VO₂, improves the dispersion of VO₂ NPs in the polymer matrix, [15] and is expected to be applied in the design of solid-state tunable F-P filters.

In this paper, based on the F-P cavity interference principle, we use VO₂/P4VP films to simulate near-infrared solid-state tunable/switchable single-/dual-band bandpass filters. The tunable single/dual passband filters exhibit excellent narrowband transmission filtering performance, with the former having a thermal tuning range of 19 nm, a half-peak width of 28.8 nm and a peak transmittance of >30% at 1064 nm, and the latter possessing thermal tuning ranges of 15 nm at 945 nm and 24 nm at 1168 nm, with a stopband width of ~100 nm, and peak transmittances of greater than 35%. On the basis of tunability, by increasing the VO₂ volume fraction of the F-P cavity and using the high extinction coefficient after the VO₂ phase transition, the switchability of the single-/dual-band bandpass filters is realized. For the first time, our simulation introduces VO₂/polymer films into the optical filter design, and uses the working principle of F-P cavity to realize NIR filters with different functions, which is of guiding significance for the design of NIR tunable filters in the future.

2. Theoretical calculation

For optical coherence tomography (OCT) system applications, the filter was chosen to be tuned in the band around 1064 nm (λ_0) [16,17]. The simulated F-P filter consists of a distributed Bragg reflector (DBR) and an intermediate refractive index change layer (VO₂/P4VP) (as shown in Fig. 1). DBR is arranged by overlapping TiO₂ and SiO₂, and the optical thickness of each layer is 1/4 of the central wavelength (λ_0). F-P cavity can achieve the selection of transmitted central wavelength λ [18]:

$$m\lambda = 2nL \cos \theta \quad (1)$$

where n , L are the refractive index and length of the F-P cavity, respectively, θ is the angle of incidence and m is the interference order of the transmitted light. From Eq. (1), it can be seen that solid-state tunable F-P filters can be achieved by varying n .

To simulate a solid-state F-P tunable filter, we chose a VO₂/P4VP films as the F-P cavity. Because P4VP can provide good dispersion of commercially available VO₂ particles (particle size of 30-50 nm), [15] the refractive index n and extinction coefficient k of VO₂/P4VP films can be calculated by [13,19,20]:

$$n_{mixture} = \varphi_{VO_2} n_{VO_2} + \varphi_{P4VP} n_{P4VP} \quad (2)$$

$$k_{mixture} = \varphi_{VO_2} k_{VO_2} + \varphi_{P4VP} k_{P4VP} \quad (3)$$

where φ_{VO_2} and φ_{P4VP} are the volume fractions of the VO₂ particles and P4VP, respectively. The refractive index of VO₂ at different temperatures is derived from the refractiveindex.info database [21]. The data was obtained by Oguntoye et al. using a RC2 ellipsometer (J.A. Woolam) installed with a heating stage to measure and fit the variable angle spectroscopic ellipsometry, [22] while the refractive index of P4VP was measured and fitted by us on the ellipsometer.

As a transmissive narrowband bandpass filter, the peak transmittance (T_{max}), free spectral range (FSR), fineness (F) and full width at half maxima (FWHM) of the filter are the key parameters to evaluate the performance of the filter, where the peak transmittance T_{max} can be expressed as:

$$T_{max} = \frac{1 - R - A}{1 - R + \alpha} \quad (4)$$

where R is the reflectivity of the mirror, A is the absorption and scattering factor of the mirror, and α is the loss in the cavity (including the loss of light caused by the absorption, scattering,

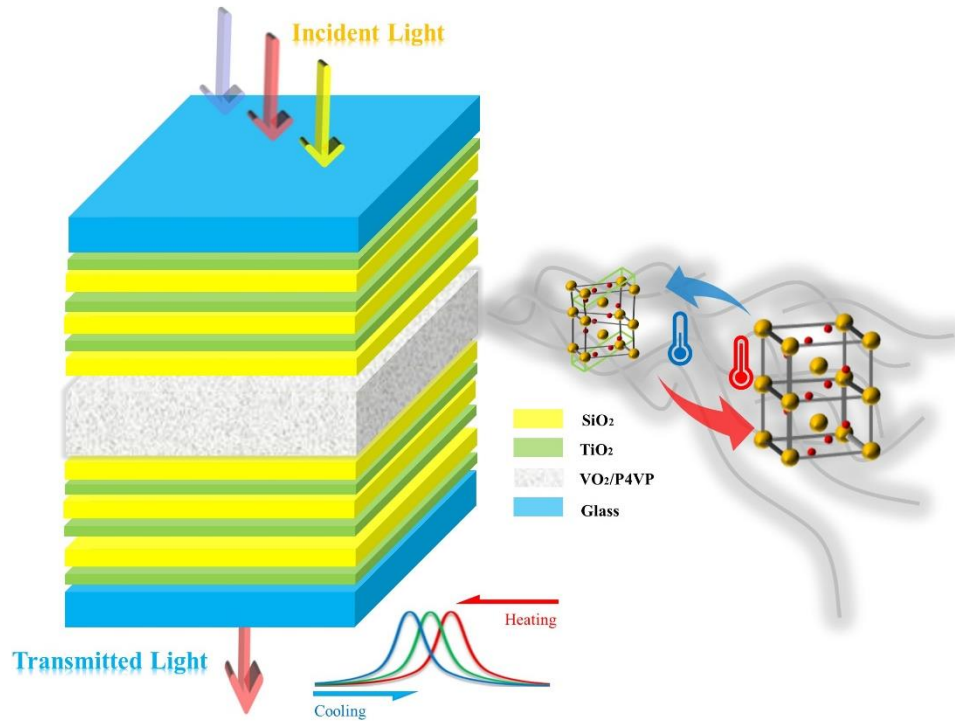


Fig. 1. Schematic diagram of the structure of the target F-P filter with three pairs of DBRs.

and bending of the F-P cavity). FSR is given through [23]:

$$FSR = \frac{\lambda_0^2}{2n_{mixture}L} \quad (5)$$

F can be expressed as [24]:

$$F = \frac{\pi\sqrt{R}}{1-R} \quad (6)$$

The FWHM can be obtained from the transmission spectrum or from [25]:

$$FWHM = \frac{FSR}{F} \quad (7)$$

These theories are valid for the F-P filter with constant reflectance mirrors around λ_0 designed in this paper, so they are used to evaluate the performance and analyze the mechanism of the filter.

3. Results and discussion

In order to obtain the refractive index of P4VP, we prepared a P4VP film with a thickness of ~500 nm using glacial acetic acid as the solvent, and then measured the refractive index of the film at 800 nm-1600 nm with an ellipsometer of about 1.51, which is close to the refractive index of PVP [21]. In addition, the fitting results showed that the extinction coefficient of the film was 0, indicating that P4VP could reduce the extinction coefficient of the F-P cavity layer and increase the peak transmittance. However, the known refractive index and extinction coefficient of VO₂ vary significantly in the near-infrared range, [26,27] which cannot be ignored in the calculation

of the F-P physical model, so when using Eqs. (2) and (3) to calculate the n , k of the mixed film, the wavelength-dependent values of n_{VO_2} and k_{VO_2} are used.

The parameters of the cavity layer are crucial to the performance of the filter. We calculated the n , k distribution of 1%-10% VO_2 volume fraction composite films at 20 °C/80 °C, and the results are shown in Fig. 2. It can be seen that with the decrease of VO_2 volume fraction in the composite film, the n , k values tend to be close to n_{P4VP} and 0. Therefore, the combination with P4VP leads to a significant decrease in the extinction coefficient of the composite films, which makes the simulation of the VO_2 -based tunable F-P filter with high transmittance feasible, and the abrupt change of the extinction coefficient at high and low temperatures makes the simulation of the VO_2 -based switchable filter feasible.

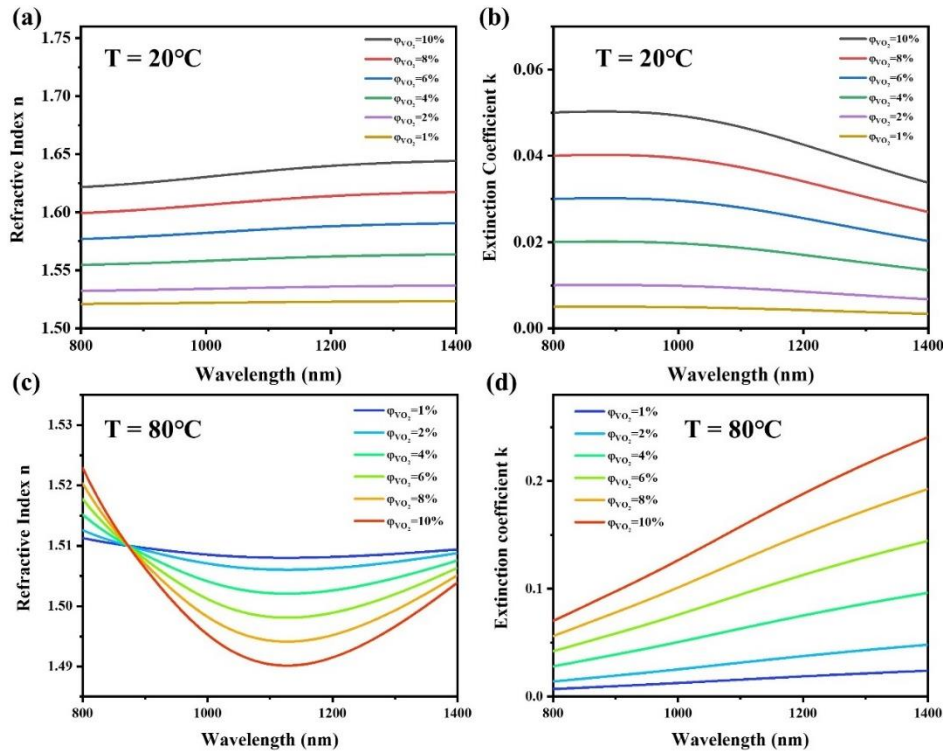


Fig. 2. Refractive index n distribution of composite films with different VO_2 volume fractions at 20 °C and c) 80 °C; Extinction coefficient k distribution of composite films with different VO_2 volume fractions at b) 20 °C and d) 80 °C.

Using COMSOL to simulate the actual VO_2 /P4VP film thickness and ensure the filter tuning range, we let Eq. (1) $m = 2$, [18] i.e., the optical length of the cavity layer at resonance is a central wavelength λ_0 (perpendicular incidence). We build an ideal 2D model of the F-P filter, as shown in Fig. 3. When we designed DBR, the near-IR refractive indices of SiO_2 and TiO_2 obtained from TFCalc fitting were 1.434 and 2.299, which are very similar to the results provided by Méndez et al [38]. and Siefke et al [39]. in the refractive index database [21]. To make it easier to observe the electric field mode distribution of the filter, we set the substrate thickness to 1 μm , the filter width to 100 nm, and the electric mode field intensity to 1 V/m in the z -axis direction and 0 in the rest direction. In order to ensure sufficiently accurate results from the finite element analysis model in this paper, we have set the finite element mesh size to a maximum cell size of $\lambda_0/n_{TiO_2}/5$. Then, we fill the n , k distributions of different composite films at room temperature

into the cavity material in the form of interpolation functions, in which the interpolation form is cubic spline and the extrapolation form is constant.

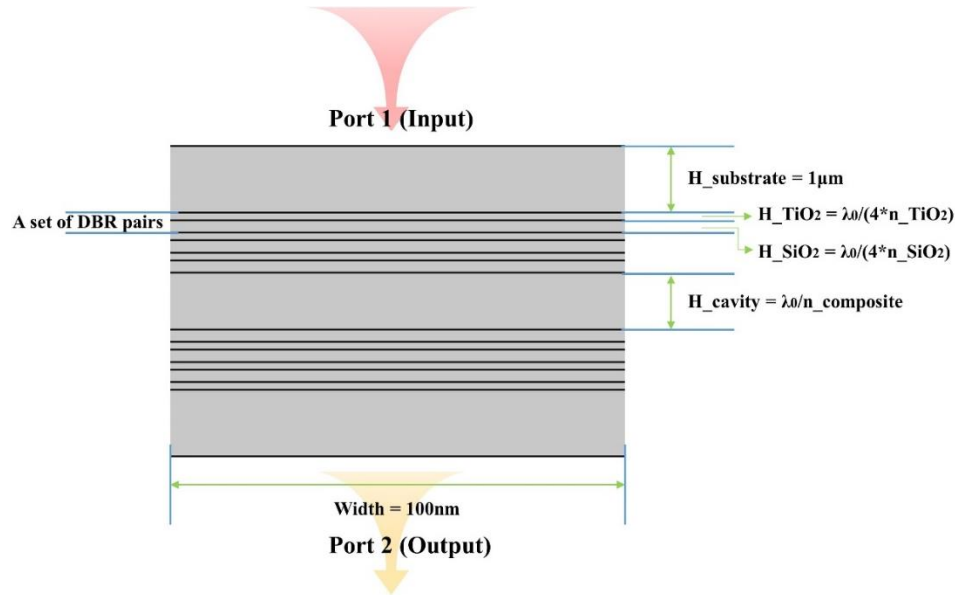


Fig. 3. COMSOL geometric model of target F-P filter.

The center wavelength of the filter is set to 1064 nm (i.e., $n_{\text{composite}} \cdot L = 1064$ nm, $n_{\text{composite}}$ is the refractive index of the cavity material at 1064 nm), the port input power is 1 W/m, and the calculated wavelength domain is 800-1400 nm. In order to observe the shift of the center wavelength in detail, the wavelength domain 1059-1069 was calculated in steps of 0.01 nm. Finally, as the number of DBR pairs and the volume fraction of VO_2 in the cavity layer varied, different filter filtering performances were observed (as shown in Fig. 4). As can be seen from the figure, with the increase of VO_2 content, the T_{max} decreases significantly, and the FWHM increases. The change in transmittance can be attributed to the increase in cavity absorption, because according to Eq. (4), an increase in VO_2 content will increase the α and thus decrease in T_{max} , while the change in FWHM is due to the fact that the increased absorption has a greater effect on the resonant light than on the light on the stop band, which ultimately leads to a larger FWHM, which is different from the trend described in Eq. (7), indicating that the intracavity loss is also a factor affecting the FWHM of the F-P filter. With the increase of the Number, the transmittance of the resonant peak decreases, and the FWHM narrows. This is because an increase in R leads to a decrease in T_{max} (Eq. (4)) and an increase in F , which in turn leads to a narrowing of the FWHM, i.e., an increase in the filter's frequency selection ability.

In order to observe the changes in T_{max} , FWHM, and peak position in more detail, we summarized the distribution of the corresponding parameters according to Fig. 4, as shown in Fig. 5. As can be seen from Fig. 5(a), the higher the volume fraction of VO_2 , the slower the rate at which the T_{max} decreases with the increase of the number of DBR pairs, and the effective transmittance can be obtained at Number = 2, 3, 4. When Number ≥ 2 , the FWHM narrows rapidly, but when Number ≥ 5 , the filter with high VO_2 content leads to low T_{max} due to high absorption, and the FWHM is not of any significance to be discussed. In addition, it can be seen from the peak position change diagram of the transmission spectrum that the increase of the Number is conducive to the stability of the center wavelength, which is due to the selection of the center wavelength by the high reflectivity of DBR.

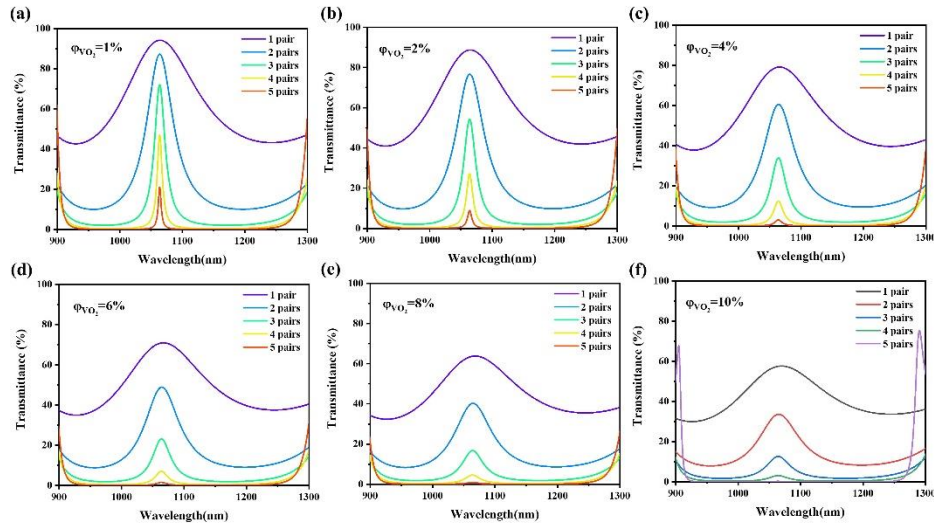


Fig. 4. Transmission spectra of F-P filters with VO₂ volume fractions of a) 1%, b) 2%, c) 4%, d) 6%, e) 8% and f) 10% at 900 nm-1300 nm.

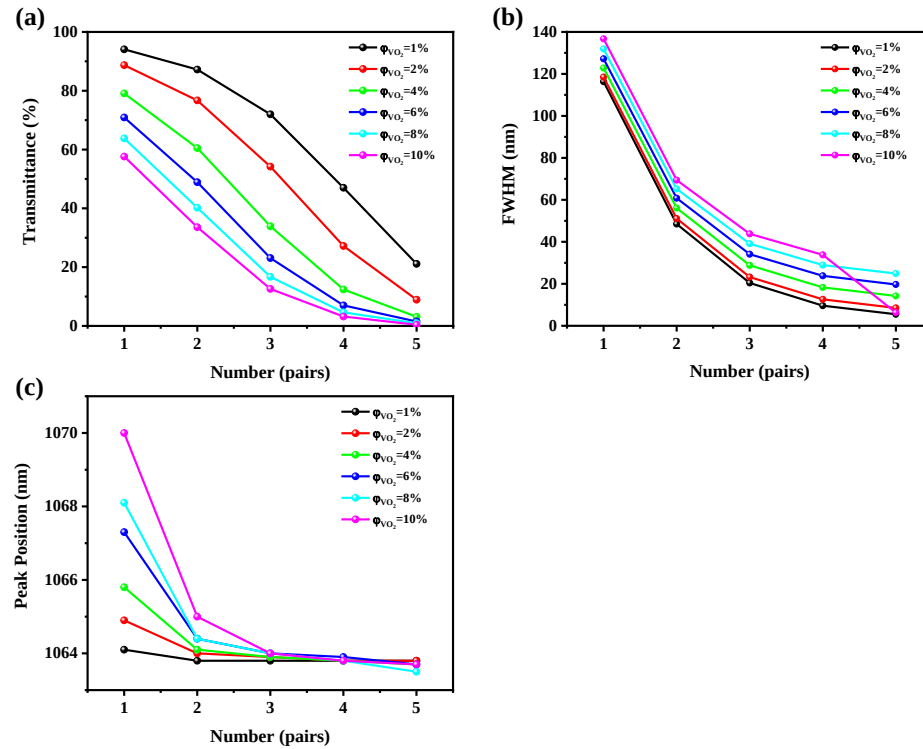


Fig. 5. a) Peak transmittance, b) FWHM, and c) peak position of filters with different VO₂ volume fractions at DBR numbers of 1, 2, 3, 4 and 5.

Based on the above analysis, we decided to calculate the transmission spectra of filters with different VO₂ content at 80 °C using a DBR with a Number between 2 and 3, and compare them with the spectral lines at 20 °C to observe the thermal tuning properties of the device, as shown in

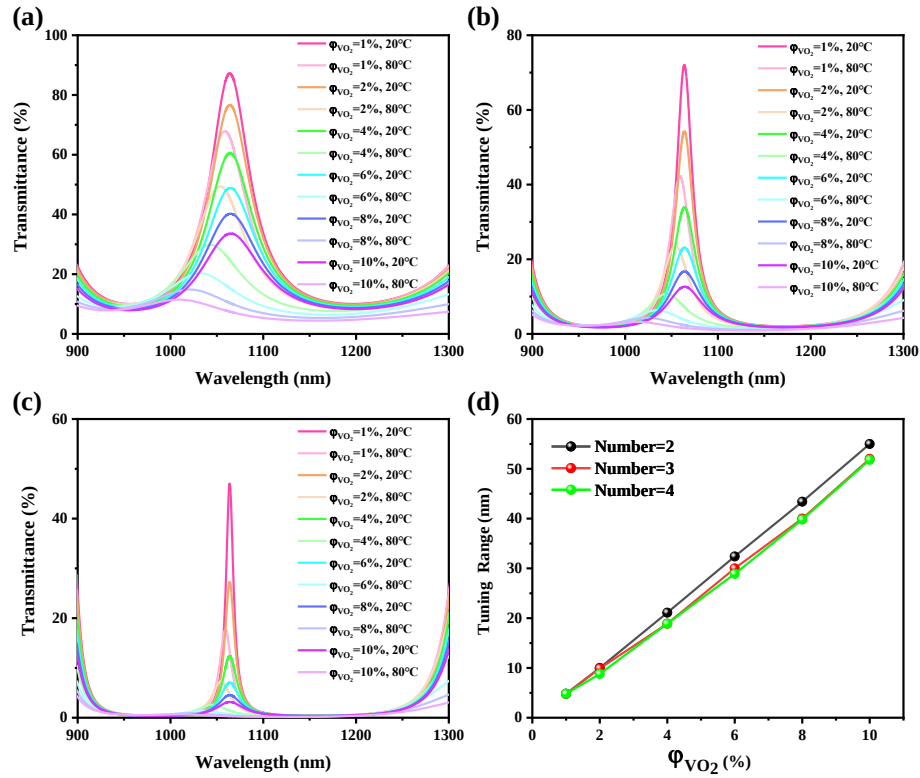


Fig. 6. High and low temperature transmission spectra of filters with different VO₂ volume fractions at DBR pair numbers a) 2, b) 3 and c) 4; d) The tuning range of filters with different parameters.

Fig. 6. At 80 °C, the transmission peak is significantly blue-shifted, which is due to the decrease of the refractive index of VO₂ at high temperature, resulting in a smaller resonance wavelength due to the F-P resonance condition. In addition, the peak transmittance decreases rapidly as the temperature increases, due to the fact that VO₂ exhibits a higher k at high temperature in the metallic phase, which increases the absorption of light by the device, as shown in Fig. 5(a). In particular, when the VO₂ volume fraction $\geq 8\%$, the transmission peak at 80 °C almost disappears, indicating that the filter can achieve single-passband filtering that can be turned off.

Figure 6(d) shows the relationship between the resonant peak offset range and the mirror and VO₂ content, and it can be seen that the tuning range of the device has no obvious relationship with the Number of mirrors, but has an obvious linear relationship with the volume fraction of VO₂, which is due to the fact that the tuning power comes entirely from VO₂, and the resonant peak offset caused by the VO₂ of the equal volume fraction is proportional and therefore positively correlated.

On the basis of the preliminary calculation of the thermal tuning results, we selected a 4% VO₂ filter with a large tuning range and considerable T_{max} to simulate the continuous thermal tuning spectrum. We sequentially fill in the cavity layer field of the COMSOL model material section with the refractive index n and extinction coefficient k of a 4% VO₂ composite films at 30-70 °C (step size is 10 °C), and then calculate the corresponding transmission spectra in stages for statistical analysis (Fig. 7(a)). When the temperature reaches 20 °C-50 °C, the transmission spectrum is basically unchanged, when the temperature is 60 °C, the transmission peak is shifted, indicating that there is a VO₂ metal phase at this time, when the temperature reaches 70 °C, the

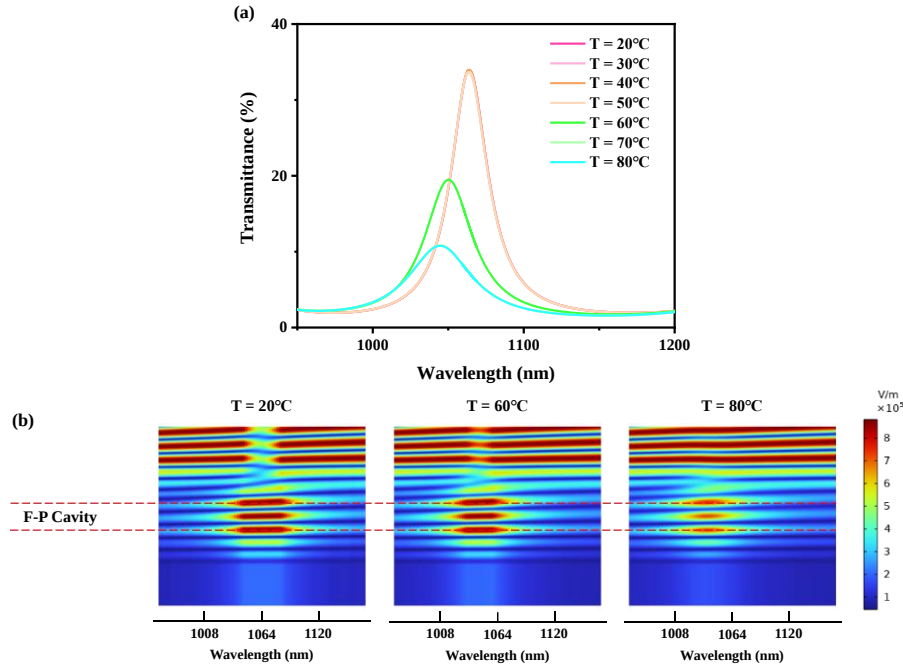


Fig. 7. a) Transmission spectra of a filter with a VO₂ volume fraction of 4% at 20 °C -80 °C. b) E-field mode distribution of the F-P filter in a) at 20 °C, 60 °C, and 80 °C.

transmission spectrum continues to be blue-shifted, and at 80 °C, the peak position is unchanged, indicating that at 70 °C, VO₂ has completely transformed into a metal phase, which is in line with the VO₂ phase transition law. Although the transmission is significantly reduced due to absorption, it also compensates for the poor filtering caused by the peak shift compared to the FWHM, and improves the filtering ability.

We then simulated the wavelength-dependent electric field mode distribution of the device at $T = 20, 60,$ and 80 °C, as shown in Fig. 7(b). At $T = 20$ °C, the F-P cavity has a strong electric field distribution around 1064 nm, which is caused by the interference of light, which is consistent with the theory. As the temperature increases, the refractive index decreases, the absorption increases, and the resonant wavelength is blue-shifted, so the strong electric field distribution appears at smaller wavelengths, and the absorption of resonant light increases and the electric field distribution weakens.

According to the interference conditions of the F-P cavity, changing the thickness of the resonator can also effectively achieve the shift of resonant peak [40,41]. Therefore, on the basis of 4% VO₂, $T = 20$ °C, $\lambda_0 = 1064$ nm filter, we changed the thickness of the cavity layer and observed the spectral trend. As shown in Fig. 8(a), as the thickness of the cavity increases, the transmission peak shifts to the right, when the thickness increases to 340 nm, a new transmission peak appears at 1064 nm, and finally the effective tuning range of the device by increasing the thickness is 200 nm. It is worth mentioning that when the thickness is increased to 200 nm, there is a double pass band between 900 nm and 1400 nm, and the transmittance can be maintained at about 30%.

To further optimize the two-pass band, we continued to adjust the thickness and obtained a more symmetrical spectrum at +140 nm, as shown in Fig. 8(b), where there is a ~ 100 nm stop band between the two resonant peaks (945 nm and 1168 nm) and $T_{max} > 35\%$, showing good dual-pass filtering performance. Combined with the characteristics of the refractive index of the

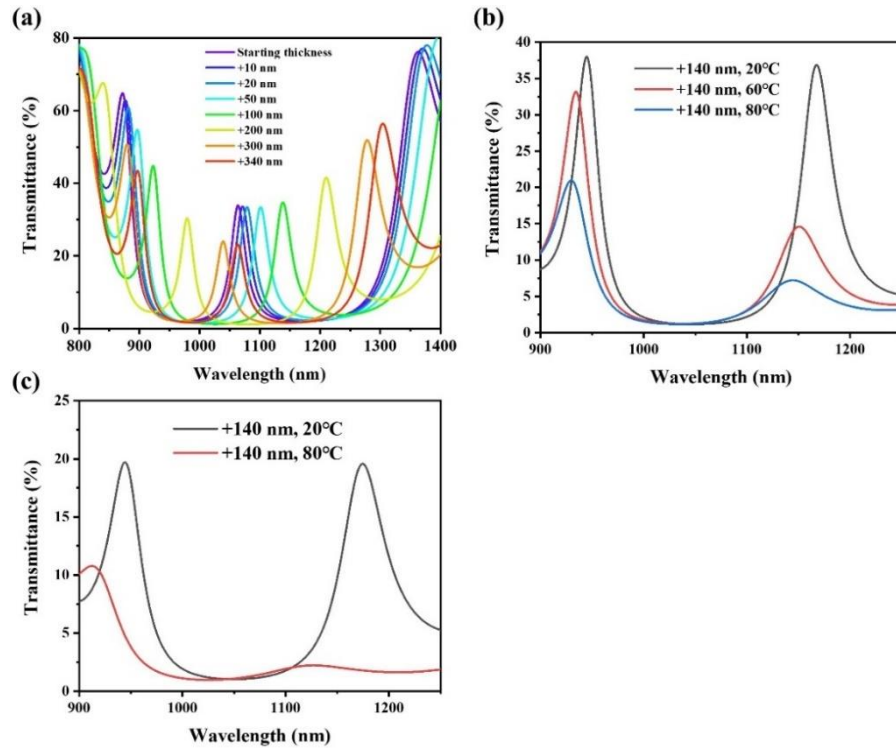


Fig. 8. a) Transmission spectra of 4% VO₂ volume fraction filter with a central wavelength of 1064 nm after increasing different cavity lengths at 20 °C; b) Thermally tuned transmission spectra of the dual-band bandpass filter formed by adding 140 nm in a); c) High- (80 °C) and low-temperature (20 °C) transmission spectra of a filter ($\varphi_{VO_2} = 8\%$) with a central wavelength of 1064 nm with a 140 nm increase in cavity length.

cavity layer as a function of temperature, we obtained the tunable dual-band bandpass spectrum at $T = 60$ and 80°C , and the continuous tuning range of the left passband was 15 nm and the right passband was 24 nm, showing good dual-band bandpass tunability. In addition, according to the above experience in designing a single-passband filter that can be turned off, we designed a dual-passband filter that can be turned off, in order to ensure that the transmittance of the transmission peak is large enough, we choose a composite film of 8% VO₂ as the cavity layer, the results are shown in Fig. 8(c), it can be seen that the high absorption of VO₂ in the infrared makes the dual-peak disappear, and a wide stop band is formed at 900-1400 nm, and the band between 1150 nm and 1200 nm is turned off. Finally, the tunable/switchable and single/dual-band passband filters we simulate have strong reference value and guiding significance in the field of near-infrared bandpass filtering.

4. Conclusion

In this paper, we design near-infrared solid-state tunable/switchable single-band/dual-band narrowband bandpass filters based on VO₂/P4VP films by using numerical simulation methods and combining with the working principle of F-P filter. The tunable single-band filter achieves a tuning range of 19 nm with only 4% volume fraction of VO₂ and the T_{max} of $> 30\%$. Tunable dual-band filters with $> 35\%$ T_{max} at 945 nm and 1168 nm, the former with 15 nm tunability and the latter achieving a 24 nm tuning range. In the case of solid-state tunable F-P filters, which are predominantly reflective, [8,36,42] the available and substantial transmittance [9,43] shown

by single- and dual-band filters is attractive. On the basis of tunability, we increase the volume fraction of VO₂, and use the high extinction coefficient of VO₂ to make the above initial resonant peak disappear, forming a large stopband coverage, and achieving good filtering switchability. In addition, on the basis of designing various filters, we analyze the influence of the resonator on the transmission spectrum of the filters, especially the key role of intracavity absorption. Our work is instructive for the design of tunable solid-state F-P filters and near-infrared narrowband bandpass filters.

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Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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